

title: Hummingbird Lab 3 - PD Control (Simulation) date: 2025-10-31 header-includes:

- `\usepackage{amsmath}`
- `\usepackage{amssymb}`
- `\usepackage{siunitx}`

Overview

- This note summarizes the PD designs used in Lab 3 for the hummingbird simulation and records the specific formulas used to compute gains from time-domain specs. All math is LaTeX to render correctly when exported to PDF (tested with pandoc and typical VS Code PDF exporters that support MathJax/KaTeX).

Models

- Longitudinal (about $\theta_e = 0$):

$$\ddot{\theta}(t) \text{ amp; } = b_\theta, \tilde{F}(t), \quad b_\theta \text{ amp; } = \frac{\ell_T}{m_1 \ell_1^2 + m_2 \ell_2^2 + J_{1y} + J_{2y}}, \quad F_e \text{ amp; } = \frac{(m_1 \ell_1 + m_2 \ell_2)g}{\ell_T}.$$

- Lateral (successive loop closure, inner roll faster than outer yaw):

$$\phi''(t) \text{ amp; } = \frac{1}{J_{1x}}, \tau(t), \quad \psi''(t) \text{ amp; } \approx b_\psi, \phi(t), \quad b_\psi \approx \frac{\ell_T F_e}{J_{zz,eq}}, \quad J_{zz,eq} \text{ amp; } = m_1 \ell_1^2 + m_2 \ell_2^2 + J_{2z} + J_{1z} + m_3(\ell_{3x}^2 + \ell_3^2)$$

Dirty Derivative (First-Order Filtered Difference)

For any signal x :

$$\hat{\dot{x}}[k] \text{ amp; } = \alpha, \hat{x}[k-1] + (1-\alpha), \frac{x[k] - x[k-1]}{T_s}, \quad \alpha \text{ amp; } = \frac{\tau_d}{\tau_d + T_s}.$$

Gain Mapping from Specs

With desired ζ and rise time t_r :

$$\omega_n \approx \frac{2.2}{t_r}.$$

- Longitudinal (pitch):

$$k_{P_\theta} = \frac{\omega_{n\theta}^2}{b_\theta}, \quad k_{D_\theta} = \frac{2\zeta_\theta \omega_{n\theta}}{b_\theta}.$$

- Inner roll:

$$k_{P_\phi} = \omega_{n\phi}^2 J_{1x}, \quad k_{D_\phi} = 2\zeta_\phi \omega_{n\phi} J_{1x}.$$

- Outer yaw:

$$k_{P_\psi} = \frac{\omega_{n\psi}^2}{b_\psi}, \quad k_{D_\psi} = \frac{2\zeta_\psi \omega_{n\psi}}{b_\psi}.$$

Bandwidth Separation

Choose inner-loop roll at least one order faster than yaw:

$$\omega_{n\phi} = M \omega_{n\psi}, \quad M \in [10, 20].$$

Control Laws (PD Only)

- Longitudinal PD (force):

$$e_\theta \text{ amp; } = \theta_d - \theta, \quad \hat{e} < em > \theta \text{ amp; } = -, \hat{\theta}, \quad \tilde{F} \text{ amp; } = k < /em > P_\theta e_\theta + k_{D_\theta} \hat{e}_\theta, \quad F \text{ amp; } = F_e + \tilde{F}.$$

- Lateral PD (successive loops):

Outer yaw (commands roll reference):

$$e_\psi = \psi_d - \psi, \quad \hat{e} < em > \psi = -, \hat{\psi}, \quad \phi_d = k < /em > P_\psi e_\psi + k_{D_\psi} \hat{e}_\psi.$$

Inner roll (commands torque):

$$e_\phi = \phi_d - \phi, \quad \hat{e} < em > \phi = -, \hat{\phi}, \quad \tau = k < /em > P_\phi e_\phi + k_{D_\phi} \hat{e}_\phi.$$

Simulation Notes

- H.7: Step θ_d by $\pm 5^\circ$ and verify well-damped tracking. The logged force F should equal F_e plus PD action.
- H.8: Step ψ_d (e.g., 30°). The roll response should be visibly faster; the rise-time ratio should be near M .